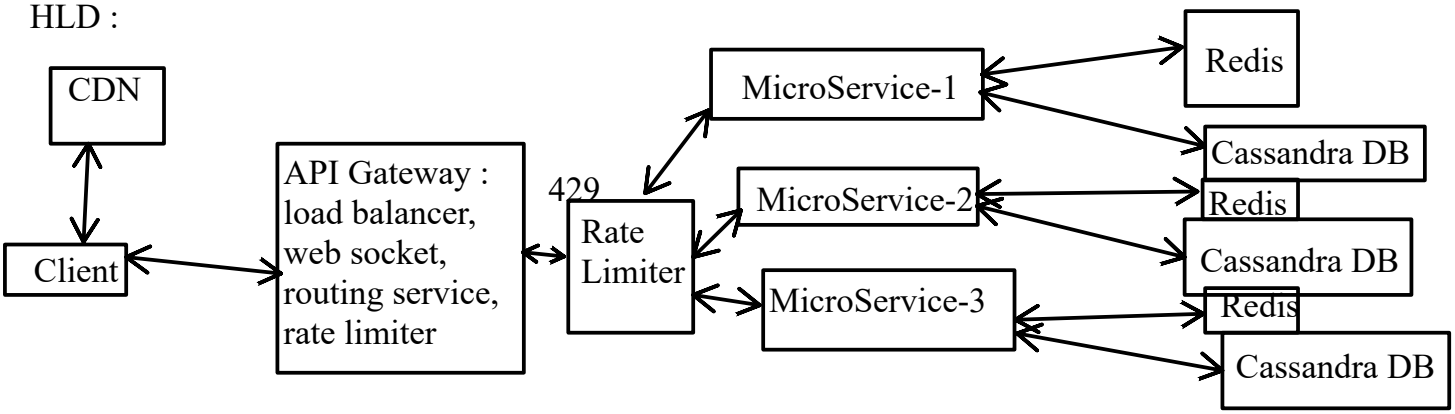
Rate Limiter

Functional requirements:
rate limiter should be able to set threshold based on the user requirements,
beyond threshold all the requests should be discarded by the rate limiter

Non-functional requirements:
Data-partitioning ,
load balancing,
fault tolerance,
vertical scalability,
horizontal sclability,
low latency,
high throughput,
CAP theorem (Here eventual consistency is fine , but here we need 99.99% uptime
i.e Availability>>Consistency)

Core Entities:
user ,
request,
history,

API :
POST:/request
{
userID,
{filemetadata}
}

HLD :